V2.4.0 • CODENAME: THE DEVOURER

TECHNICAL PROJECT ABSTRACT & ARCHITECTURE SPECIFICATION

DEVMIND AI — SINGULARITY: EXTREME

Production-Hardened Full-Stack AI Repository Intelligence Workstation & Real-Time Relativistic Kerr Black-Hole Simulation Platform

Frontend: React 18, TypeScript, Three.js / R3F, Custom GLSL, Zustand
Backend: FastAPI, SQLAlchemy 2.x, PostgreSQL + pgvector / SQLite
Audio: 100% Procedural Web Audio API (6 Layers)
AI & Security: Google Gemini, Firebase OIDC, Argon2id RBAC

1. EXECUTIVE ABSTRACT

DevMind AI — SINGULARITY: EXTREME (v2.4 — THE DEVOURER) is a full-stack artificial intelligence software engineering workstation integrated inside a real-time 3D relativistic Kerr black-hole cosmic simulation. Modern software teams struggle to navigate complex multi-language repositories, audit code for Common Weakness Enumerations (CWEs), defend LLM workflows against adversarial prompt injection, and retain architectural context across sessions. DevMind AI addresses these challenges through ten specialized intelligence modules powered by **Hybrid Vector Retrieval-Augmented Generation (RAG)**, Abstract Syntax Tree (AST) symbol indexing, and **Google Gemini** synthesis, all surfaced through a crystal-glass spatial workspace.

Surrounding the workstation is a deterministic **120 Hz astrophysical simulation and custom GLSL shader engine** inspired by European Southern Observatory (ESO) and NASA Goddard supercomputer visualizations (`AT2019qiz` , `ASASSN-14li` , `SVS 14620`). The engine renders a near-extremal rotating Kerr black hole ($a_* \approx 0.94$) featuring a depth-occluding D-shaped Event Horizon shadow, nested relativistic photon rings (n = 1, 2, 3), Einstein gravitational starfield lensing, multi-stratum volumetric accretion gas, sequential 6-stage stellar Tidal Disruption Event (TDE) spaghettification, and planetary Roche limit fragmentation—synchronized with a zero-dependency, **6-layer procedural Web Audio API soundscape** and hardened by enterprise-grade **OAuth 2.0 / OIDC + Argon2id Role-Based Access Control (RBAC)**.

Keywords: Repository Intelligence, Hybrid Vector RAG, Abstract Syntax Tree (AST), Google Gemini, Relativistic Kerr Black Hole, Tidal Disruption Event (TDE), Procedural Web Audio Synthesis, FastAPI, PromptShield, Defense-in-Depth Security.

2. CORE ARCHITECTURAL PILLARS

01 AI Repository Intelligence Workstation

- **Singularity Codebase Chat:** Hybrid vector + lexical RAG with line-level AST symbol citations and multi-turn Google Gemini synthesis.
- **Secure Repository Indexer:** Multi-file & `.zip` archive ingestion with AST parsing, language breakdown, and vector chunking.
- **PromptShield & AI Code Review:** Pre-flight adversarial prompt injection firewall paired with static CWE vulnerability scanning and automated PR review.
- **Orbital Explorer, MemoryMesh & ModelDoctor:** Interactive 3D dependency topology, semantic developer knowledge graph, live LLM latency/token calibration probes, and AST Markdown documentation generator.

02 Relativistic Kerr Black-Hole Visual Engine

- **Kerr Shadow & Photon Rings:** Pure `#000000` D-flattened shadow (`EventHorizon.tsx`) with nested n = 1, 2, 3 relativistic photon rings and upper/lower lensed disk arches (`GravitationalLens.tsx`).
- **4-Layer Volumetric Accretion Structure:** Blazing ISCO plasma ring, filamentary MHD-inspired main disk, 3D flared corona strata, and outer cooling dust torus with Doppler beaming & filmic HDR rolloff.
- **Stellar TDE & Planetary Roche Physics:** `120 Hz` fixed-step simulation driving 6-stage stellar spaghettification (44-segment bound fallback ribbon, pericenter shock hotspot, unbound outflow tail) and 3D planetary boulder fragmentation.
- **Gravitational UI & Text Physics:** Trajectory-verified glyph ingestion (`GravityTextEffect`) with native `Ctrl+Z` / `Ctrl+Y` undo reversal and full-viewport `CosmicShockwaveOverlay` .

03 Procedural Web Audio API Cosmic Engine

- **100% Mathematical Synthesis:** Zero external audio files; generates real-time audio across 6 independent layers: *The Void Drone*, *Gravitational Pressure*, *Accretion Storm*, *Stellar Destruction*, *Planetary Destruction*, and *Cosmic Shockwave*.
- **Mastering & Acoustic Safety:** Routed through a `28 Hz` subsonic highpass, `6.5 kHz` lowpass ceiling, `2.2s` dark convolver reverb, and `-4.0 dB` (`10:1`) soft-knee limiter across 5 live presets (including `SINGULARITY` & `CINEMATIC_HORROR`).
- **Autoplay & Visual Sync:** Starts strictly silent until user gesture, mutes on unauthenticated screens, and drives zero-allocation `AnalyserNode` 3D visual modulation.

04 Defense-in-Depth Security & RBAC

- **Dual Authentication & Revocation:** Official Firebase Google OAuth 2.0 / OIDC (RS256 ID token verification) + `Argon2id` password hashing and `token_version` session revocation.
- **Strict RBAC & IDOR Isolation:** Server-enforced `user` / `admin` roles (`require_admin`), `get_owned_repo()` per-user isolation across all endpoints, and request-ID structured audit logging.
- **Upload & Network Hardening:** ZIP-slip traversal guard, symlink rejection, `50x` compression bomb cap, per-IP/user sliding-window rate limiting (`429 Retry-After`), and strict CSP/HSTS headers.

3. FULL-STACK SUBSYSTEM & MODULE MATRIX

Subsystem Layer	Key Source Modules	Technical Implementation & Capabilities
3D Relativistic & Celestial Engine	`src/engine/CosmicSimulationManager.ts` `components/space/EventHorizon.tsx` `StarCaptureSystem.tsx` , `PlanetCaptureSystem.tsx`	`120 Hz` fixed-timestep integrator with linear render interpolation; zero per-frame heap allocation; 4 stellar spectral classes & 5 planetary archetypes; `PERFORMANCE` , `BALANCED` , and `CINEMATIC` quality tiers.
Procedural Cosmic Audio Subsystem	`src/audio/CosmicAudioEngine.ts` `CosmicSoundGenerator.ts` `AudioPerformanceManager.ts`	Single `AudioContext` lifecycle; seeded PRNG noise buffers; exact-once `particleId` & celestial event deduplication (2,048-entry ring); bounded transient polyphony with deterministic node cleanup.
Workstation UI & Glass System	`src/modules/*` (10 Modules) `components/glass/DynamicGlassLighting.tsx` `GravityTextEffect.tsx`	Crystal-clear dynamic glass (`blur(18px) saturate(165%)`) with shockwave-reactive specular borders; passive bubble-phase deletion tracking preserving native browser undo/redo history.
Backend API, RAG & Security Plane	`backend/app/main.py` , `auth.py` `security.py` , `rag_engine.py` `models.py` , `database.py`	FastAPI async REST plane; fail-closed `validate_startup_config()` ; SQLAlchemy 2.x ORM with PostgreSQL + `pgvector` (or local async SQLite); server-side Gemini key isolation.

120 Hz	10 Modules	6 Layers	0 CVEs
DETERMINISTIC PHYSICS STEP	AI ENGINEERING WORKSTATION	PROCEDURAL WEB AUDIO	PROD NPM & PIP-AUDIT